1. Consider the following set of processes with the given arrival times and CPU burst times:

Process	Arrival time (ms)	CPU burst time (ms)	I/O burst time (ms)
P1	0	5	5
P2	2	6	2
P3	3	1	5
P4	15	6	3

Apply FCFS, SJF, and SRTF scheduling algorithms for CPU separately to execute the given processes. **Assume that only one I/O device is available and it operates in FCFS manner**. Draw separate Gantt chart for each algorithm to show when each process will be scheduled. Calculate the **waiting times**, **turnaround times (tat)** and **CPU utilizations** of the processes for all algorithms. Show all intermediate steps of your calculations. For simplicity, ignore any potential context switch time.

FCFS

Gantt charts for the CPU and I/O devices:

W(P1): W(P2):

W(P3): W(P4):

tat(P1): tat(P2):

tat(P3): tat(P4):

CPU Utilization:

SJF

Gantt charts for the CPU and I/O devices:

W(P1): W(P2):

W(P3): W(P4):

tat(P1): tat(P2):

tat(P3): tat(P4):

CPU Utilization:

SRTF

Gantt charts for the CPU and I/O devices:

W(P1): W(P2):

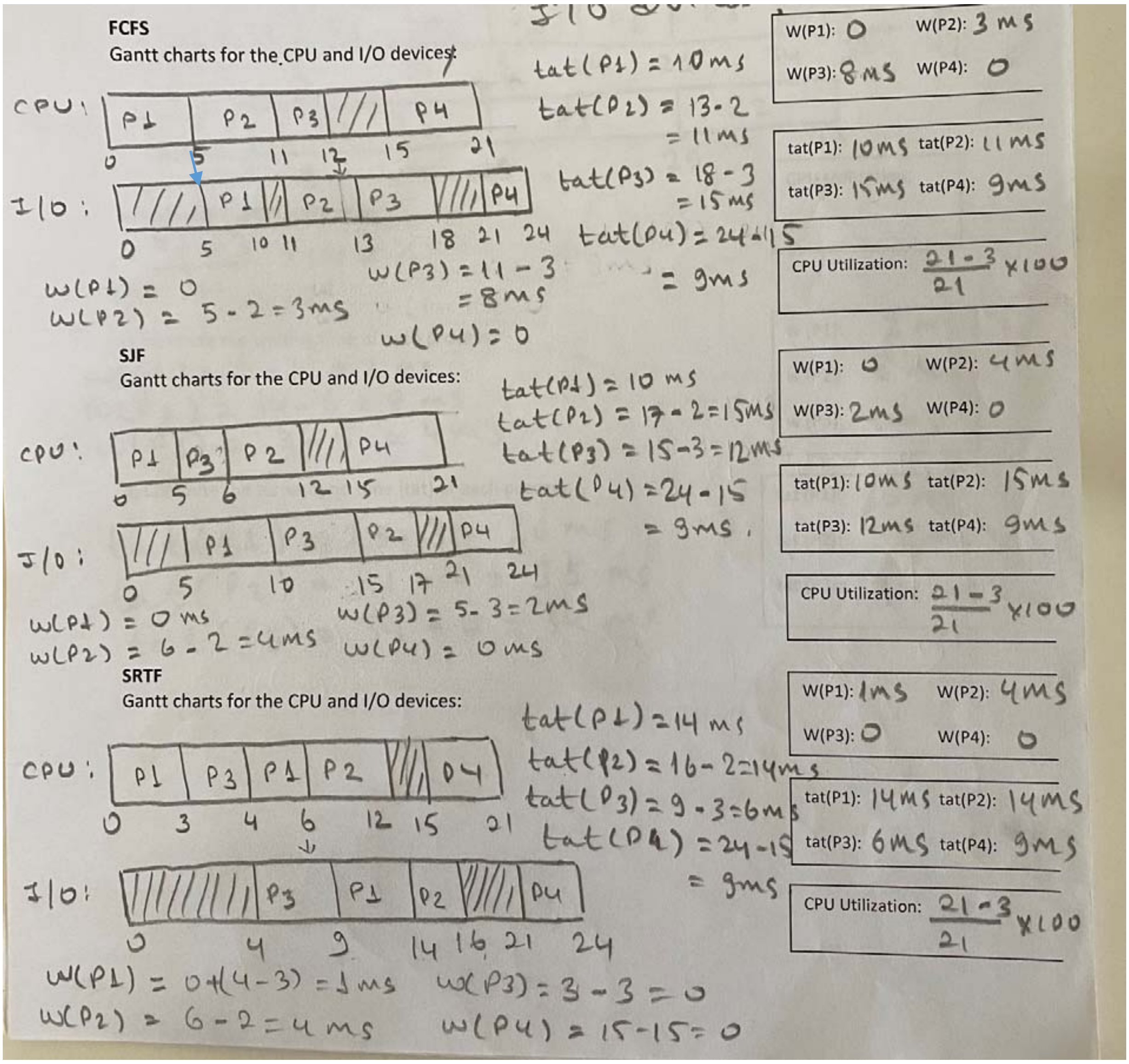
W(P3): W(P4):

tat(P1): tat(P2):

tat(P3): tat(P4):

CPU Utilization:

Process	Arrival time (ms)	CPU burst time (ms)	I/O burst time (ms)
P1	0	5	5
P2	2	6	2
P3	3	1	5
P4	15	6	3



FCFS

Gantt charts for the CPU and I/O devices:

CPU: P1 | P2 | P3 | /// | P4 — 0, 5, 11, 12, 15, 21

I/O: //// | P1 | // | P2 | P3 | //// | P4 — 0, 5, 10, 11, 13, 18, 21, 24

w(P1) = 0
w(P2) = 5 - 2 = 3ms
w(P3) = 11 - 3 = 8ms
w(P4) = 0

tat(P1) = 10ms
tat(P2) = 13 - 2 = 11ms
tat(P3) = 18 - 3 = 15ms
tat(P4) = 24 - 15 = 9ms

W(P1): 0 W(P2): 3 ms
W(P3): 8 ms W(P4): 0

tat(P1): 10ms tat(P2): 11ms
tat(P3): 15ms tat(P4): 9ms

CPU Utilization: $\frac{21-3}{21} \times 100$

SJF

Gantt charts for the CPU and I/O devices:

CPU: P1 | P3 | P2 | /// | P4 — 0, 5, 6, 12, 15, 21

I/O: /// | P1 | P3 | P2 | /// | P4 — 0, 5, 10, 15, 17, 21, 24

w(P1) = 0 ms
w(P2) = 6 - 2 = 4ms
w(P3) = 5 - 3 = 2ms
w(P4) = 0 ms

tat(P1) = 10 ms
tat(P2) = 17 - 2 = 15ms
tat(P3) = 15 - 3 = 12ms
tat(P4) = 24 - 15 = 9ms

W(P1): 0 W(P2): 4ms
W(P3): 2ms W(P4): 0

tat(P1): 10ms tat(P2): 15ms
tat(P3): 12ms tat(P4): 9ms

CPU Utilization: $\frac{21-3}{21} \times 100$

SRTF

Gantt charts for the CPU and I/O devices:

CPU: P1 | P3 | P1 | P2 | /// | P4 — 0, 3, 4, 6, 12, 15, 21

I/O: //////// | P3 | P1 | P2 | //// | P4 — 0, 4, 9, 14, 16, 21, 24

w(P1) = 0 + (4 - 3) = 1ms
w(P2) = 6 - 2 = 4 ms
w(P3) = 3 - 3 = 0
w(P4) = 15 - 15 = 0

tat(P1) = 14 ms
tat(P2) = 16 - 2 = 14ms
tat(P3) = 9 - 3 = 6ms
tat(P4) = 24 - 15 = 9ms

W(P1): 1ms W(P2): 4ms
W(P3): 0 W(P4): 0

tat(P1): 14ms tat(P2): 14ms
tat(P3): 6ms tat(P4): 9ms

CPU Utilization: $\frac{21-3}{21} \times 100$

Exercise: Compute average waiting time of processes for each algorithm.

2. Consider the following process arrival, CPU and I/O burst times. **Assume that there is only one CPU and it uses SRTF algorithm.**

Process	**Arrival time**	**CPU**	**I/O**	**CPU**
P1	0	7	6	2
P2	3	5	2	1
P3	5	1	10	6

a) **Assume that only one I/O device is available and it operates using FCFS algorithm.**

Draw the Gantt charts for the CPU and I/O device to compute the waiting time and turnaround time (tat) of all processes and CPU utilization. Put your answers into the table given below.

CPU Gantt Chart

P1	P3	P1	P2		P3	P1	P2

0 5 6 8 13 16 22 24 25

I/O Gantt Chart

	P3	P1	P2

0 6 16 22 24

W(P1)=(6-5)=1

Wait(P1)	Wait(P2)	Wait(P3)	Tat(P1)	Tat(P2)	Tat(P3)	CPU utilization
1	5	0	24	22	17	(22/25)x100%

Average waiting time = (1+5+0)/3

b) **Exercise:** Assume that only one I/O device is available and it operates using SJF algorithm.
Draw the Gantt charts for the CPU and I/O device to compute the waiting time and turnaround time (tat) of all processes and CPU utilization.

3. Consider the following process arrival, CPU, and I/O burst times. Assume you have only one I/O device and it operates FCFS manner.

Process	**Arrival time**	**CPU**	**I/O**	**CPU**
P_1	2	~~7~~ ~~6~~	~~6~~	11
P_2	0	~~12~~ ~~10~~	3	10
P_3	3	~~2~~	~~6~~	~~6~~

Using **SRTF** process scheduling algorithm:

Gantt Charts of CPU and I/O:

CPU	**P_2**	**P_1**	**P_3**	**P_1**	**P_3**	**P_2**	**P_1**	**P_2**

0 2 3 5 11 17 27 30 38 48

I/O		**P_3**	**P_1**		**P_2**

W(P1)=(5-3)+(27-17)=12 W(P2)=(17-2)+(38-30)

a) Calculate waiting times of Processes?
$W(P_1)$ =__12__ $W(P_2)$ =___23___ $W(P_3)$ =__0____

Average waiting time = (12+23+0)/3

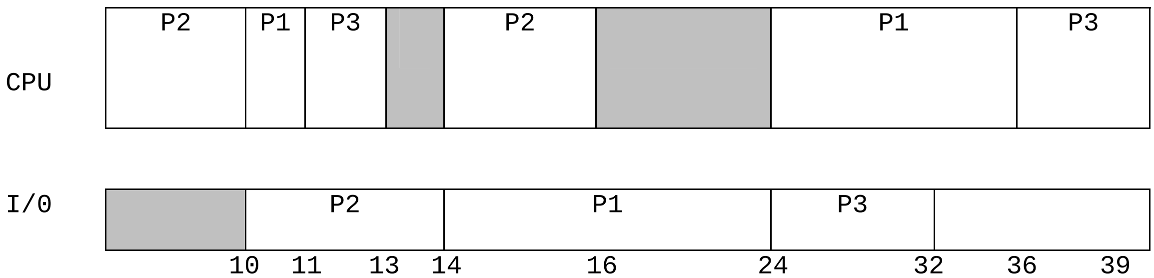
b) Calculate turnaround times of processes?

$tat(P_1)$ =__36____ $tat(P_2)$ =___48___ $tat(P_3)$ =__14____

4. Consider the following process arrival, CPU and I/O burst times. Assume that only one I/O device is available and it operates in FCFS manner. Assume that there is only one CPU and it uses **shortest job first** scheduling algorithm.

Process	Arrival time	CPU	I/O	CPU
P_1	3	1	10	12
P_2	0	10	4	2
P_3	5	2	8	3

Gantt Charts of CPU and I/O:



a) Calculate waiting times of Processes?
W(P_1) =__7____ W(P_2) =__0____ W(P_3) =__10____

Average waiting time = (7+0+10)/3

b) Calculate turnaround times of processes?
tat(P_1) =_33___ tat(P_2) =__16___ tat(P_3) =__34____

c) CPU utilization = __(30/39) X 100 % ____

5. Consider the following set of processes:

Process ID	Arrival Time	CPU Burst Time	I/O Burst Time	CPU Burst Time
P_0	2	1	1	2
P_1	1	3	7	1
P_2	0	5	2	1

a) Draw Gantt charts illustrating the execution of these processes using the Shortest Remaining Time First (SRTF) algorithm. (note: I/O device uses FCFS scheduling)

P_2	P_1	P_0	P_1	P_0	P_2		P_1		P_2
0	1	2	3	5	7	11	12	13	14 15

	P_0		P_1	P_2	
0					

b) Based on your work above, fill in the table below giving both the **waiting time (Wait)** and **turnaround time (tat)** for each process:

Scheduling Algorithm	**Parameter**	**Process ID**		
		P_0	**P_1**	**P_2**
SRTF	**Wait**	1	1	6
	Tat	5	12	15

c) Calculate the CPU utilization.

(13/15)x100

6. Given the following table, how will these processes be scheduled using FCFS, SJF and SRTF algorithms? Create a Gantt chart showing when each process will be scheduled, and calculate the waiting times and turnaround times for the processes for each scheduling algorithm.

Process	**Arrival time**	**CPU-Burst time**
$\mathbf{P_0}$	0	6
$\mathbf{P_1}$	3	2
$\mathbf{P_2}$	4	7
$\mathbf{P_3}$	7	1
$\mathbf{P_4}$	8	3

(FCFC) CPU Gantt Chart:

$\mathbf{P_0}$	$\mathbf{P_1}$	$\mathbf{P_2}$	$\mathbf{P_3}$	$\mathbf{P_4}$

0 6 8 15 16 19

(SJF) CPU Gantt Chart:

$\mathbf{P_0}$	$\mathbf{P_1}$	$\mathbf{P_3}$	$\mathbf{P_4}$	$\mathbf{P_2}$

0 6 8 15 16 19

(SRTF) CPU Gantt Chart:

$\mathbf{P_0}$	$\mathbf{P_1}$	$\mathbf{P_0}$	$\mathbf{P_3}$	$\mathbf{P_4}$	$\mathbf{P_2}$

0 3 5 8 9 12 19

	W(P_0)	**W(P_1)**	**W(P_2)**	**W(P_3)**	**W(P_4)**	**Average waiting time**
FCFS	0	3	4	8	8	(0+3+4+8+8)/5
SJF	0	3	8	1	1	(0+3+12+1+7)/5
SRTF	2	0	8	1	1	(2+0+8+1+1)/5

	tat(P_0)	**tat(P_1)**	**tat(P_2)**	**tat(P_3)**	**tat(P_4)**
FCFS	6	5	11	9	11
SJF	6	5	15	2	4
SRTF	8	2	15	2	4